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🏎️ FH6SmartShift (v1.1.1)

Virtual Dyno, Dragy Timer, Tuning Advisor & Shift Telemetry for Forza Horizon 6 & Forza Motorsport

Platform: SimHub | Games: FH6 & Forza Motorsport | .NET Framework 4.8 + WPF | License: GPLv3

User & Technical Manual
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Build: v1.1.1 Dual-Game
Sampling: 60 Hz UDP

1. BEST PRACTICE: HOW TO CALIBRATE THE ENGINE MAP (100% DYNO) & GEARS

🎯 The Golden Rule for 100% Repeatable Dyno Curves & Tuning Advice

Why a standing 0–200 km/h launch isn't enough: During a hard launch, 1st gear spins the tires (filtered out by traction gates) and upshifts into Gears 2–6 land high in the powerband (above `~4,800-5,500 RPM`), skipping the `2,500-4,800 RPM` mid-range where Peak Torque lives.

1. **Step 1 — The 3rd or 4th Gear Dyno Pull (2,500 RPM to Limiter):** Shift into **3rd or 4th gear** at low speed (below `2,500 RPM`) and hold **100% Wide-Open Throttle (WOT)** all the way to the rev limiter. A higher gear prevents wheelspin, lets turbochargers reach 100% boost early, and sweeps every `50 RPM` band up to redline — switching the dashboard badge to `DYNO OK 100%` and unlocking the HTML Dyno Chart & Tuning Advisor.
2. **Step 2 — Lock All Gear Ratios (Gears 1–10):** Accelerate cleanly through each remaining gear (1st through top gear) for ~0.4 s per gear until every slot locks Blue `(R)` and transitions to Green `(P)` (Math Peak) or Green `(L)` (Limiter Optimal).

2. CORE FEATURES (V1.0 FOUNDATION — BUILT-IN SINCE LAUNCH)

📊 401-Band Live Dyno
Maps Power, Torque & Boost every `50 RPM` (up to `20,000 RPM`) at WOT (`Accel >= 240`) with asymmetric EMA filtering and a 20-block `MapCoverageBar`.

📈 Math Shift Intersections
Calculates exact wheel-power crossover (`pNext >= 1.015 * pNow` across 5 steps) with a `0.85` turbo boost-drop penalty across Gears 1–10.

⚙️ Auto Ratio & Tune Reset
Locks mechanical ratios (`RpmPerRadS`) via 25-sample IQR filtering and auto-resets ratios when transmission slider changes are detected.

👻 Ghost Limiter Detection
Detects hidden power cuts before visual `EngineMaxRpm`, purges contaminated bands, and auto-unlocks if the engine later revs higher.

⚡ Reaction-Time Buffer
Tracks real-time RPM acceleration (`RPM/s`) and triggers shift lights early (up to `1500 RPM` in G1–2, `600 RPM` in G3+) to prevent limiter bounce.

🔽 Smart Downshift & Climb
Raises safe downshift RPM under braking (`+600 RPM`) & front dive (`+250 RPM`), plus suggests incline downshifts when lower gear gives `>= +5%` power.

🛡️ 7-Layer Physics Filter
Pauses dyno learning during wheelspin, jumps, puddles, kerbs/rumble, rear rebound spikes (`> 0.08 m`), damage, clutch, and shift cooldowns.

🏎️ Chassis & Aero Alerts
Includes `IsGripLost` (`> 1.25 G`), RWD/AWD `PowerOversteerWarning`, high-speed `AeroBottomingWarning` (`> 200 km/h`), and `BodySlipAngle`.

💾 Per-Build DB & Backups
Saves profiles by `{Ordinal}_PI_DT_Rpm`, migrates legacy profiles, saves asynchronously in background threads, and keeps 3 daily backups.

3. WHAT'S NEW IN V1.1.0 & V1.1.1 (LATEST EXPANSION)

🏁 1. Dual-Game Support (FH6 & Forza Motorsport)
Auto-detects active game (`ActiveGameMode` = `"FH6"` or `"FM"`), isolates Motorsport profiles with an `FM_` prefix, and tightens track-tire slip gates.

🔒 2. 100% Dyno Map Gate, Turbo Gate & Self-Healing Ratios
Locks HTML Dyno Chart & Tuning Advisor until `100%` map completion, rejects low-boost turbo spool transients (`< 78%` peak boost), and self-heals slip-deflated gear ratios.

📊 3. HTML Garage Inspector & Gearbox Tuning Advisor
Standalone browser report (`FH6SmartShift_GarageReport.html`) with Gears 1–10 drop/landing table, 401-band dyno canvas, and automated slider recommendations.

⏱️ 4. Sub-Frame "Dragy" Timer & F1 Audio Shift Beep
Measures **0–100 km/h**, **100–200 km/h**, and **0–200 km/h** with Personal Best persistence, plus an optional `2650 Hz` F1 Shift Beep (Default: OFF).

🏷️ 5. Custom Car Naming, Single-Car Export & DB Merge
Name cars/tunes via `✏️ Name Car` (`FH6SmartShift_CarList.json`), export single car profiles (`📤 Export Car`), or merge external JSON databases.

🖥️ 6. WPF Control Center, Active Gear & 3-Color Slide Bar
SimHub WPF tab with **Reaction Buffer Slider (0.10–0.40 s)**, `CurrentGearColor` / `Text` delegates, `LedActiveCount (1..12)`, and 3-color Slide Recovery Bar.

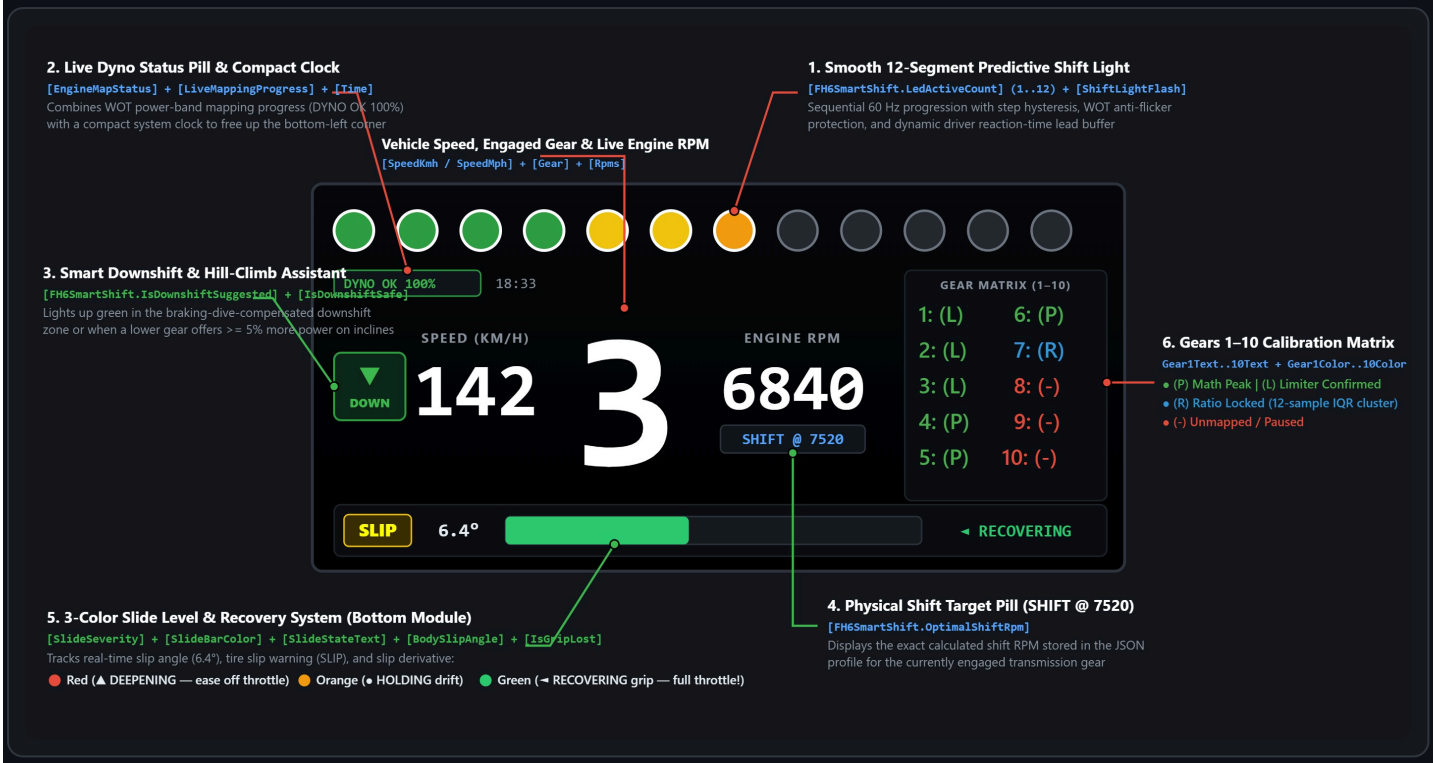
4. QUICK INSTALLATION & FILE LOCATIONS

1. Close SimHub completely and copy `FH6SmartShift.dll` into your SimHub root folder (`C:\Program Files (x86)\SimHub\`).
2. Start SimHub, enable the plugin, and import `FH6SmartShift_Optimal.zip` / `DashTemplates.zip` into Dash Studio.

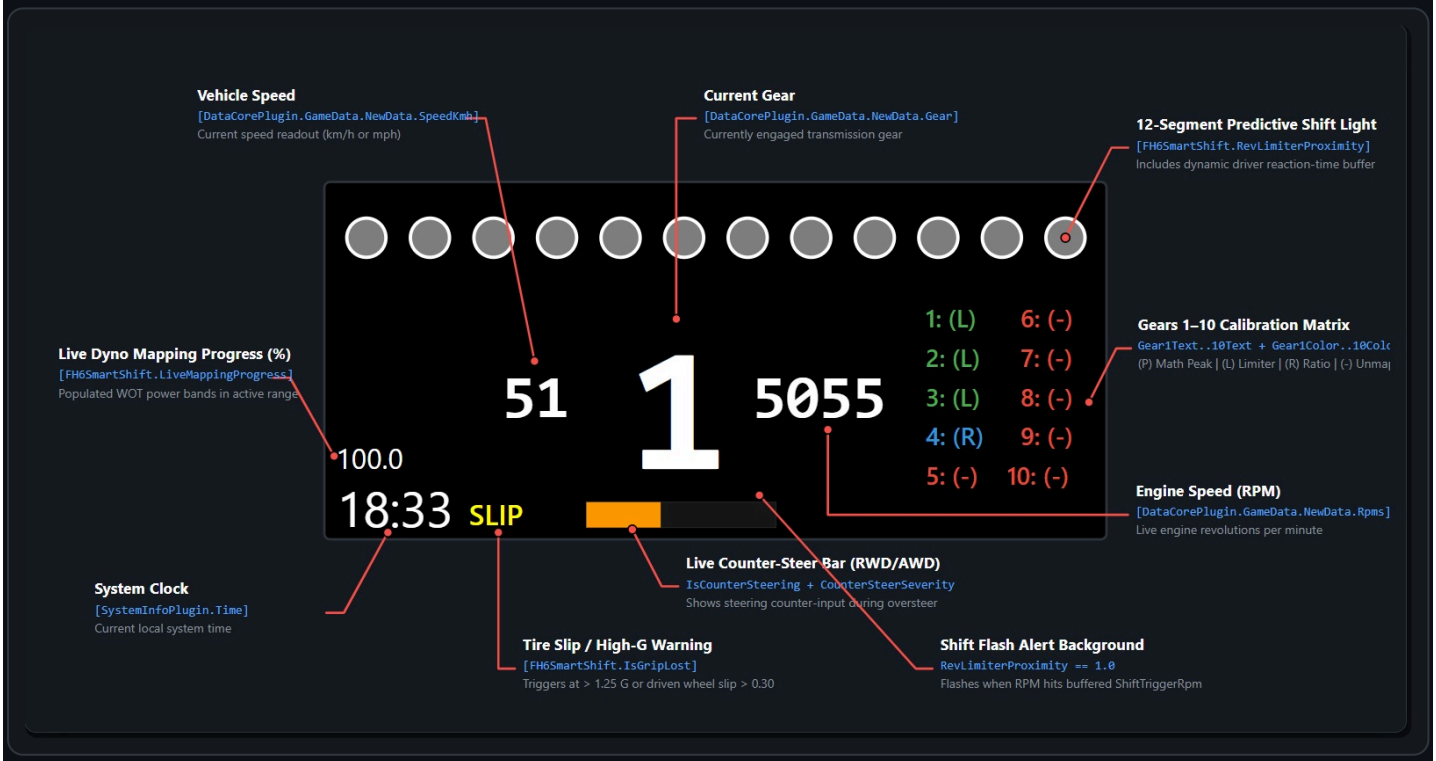
3. In **Forza Horizon 6** or **Forza Motorsport** (**HUD and Gameplay → Data Out**), enable UDP telemetry matching SimHub's IP and Port.
4. **Database, CarList & HTML Report:** Stored in `C:\Program Files (x86)\SimHub\PluginsData\Common\` (retains 3 daily backups `.bak_yyyyMMdd`).

5. DASHBOARD VISUAL REFERENCE (V1.1 OPTIMAL & CLASSIC LAYOUTS)

v1.1 Optimal MoTeC-Style Layout (1334×513 / 980×377): Features a sequential 60 Hz 12-LED bar (`LedActiveCount 1..12`), compact Dyno Status Pill (`DYNO OK 100%`), Smart Downshift Assistant (`▼ DOWN`), Live Shift Target Pill (`SHIFT @ 7520`), Multi-Warning Pill (`SLIP` / `OVRST!` / `BOTTOM!`), and the **3-Color Slide Level & Recovery Bar** (🔴 Red `▲ DEEPENING` | 🟠 Orange `● HOLDING` | 🟢 Green `◄ RECOVERING`).



Classic Telemetry Layout: Features the 12-segment predictive shift light bar, central Gear/Speed/RPM cluster, Gears 1–10 calibration matrix, and bottom slip/counter-steer indicators.



6. GEARS 1–10 CALIBRATION MATRIX STATES

State	Color / Hex	Actual Condition in C# Code (v1.1.1)
`N: (P)`	🟢 Green `#4CAF50`	**Peak Optimal (`IsConfirmed && IsMathOptimal`):** Confirmed power curve intersection with gear `N+1` (`pNext >= 1.015 * pNow` across 5 steps) and active landing zone density `>= 78%`.
`N: (L)`	🟢 Green `#4CAF50`	**Limiter Confirmed (`IsConfirmed && !IsMathOptimal`):** Confirmed optimal shift right at the true/ghost rev limiter (no power intersection before cutoff, or confirmed top gear).
`N: (L?)`	🟠 Orange `#F39C12`	**Edge / Unconfirmed Limit (`TriggerIsEdge`):** Provisional trigger (`actualLimiter - 250 RPM`) while the upper power band is still being populated.
`N: (P?)`	🟡 Yellow `#F1C40F`	**Unconfirmed Trigger:** Gear has a saved shift point (`ShiftTriggerRpm > 0`), awaiting full density confirmation.
`N: (R)`	🔵 Blue `#3498DB`	**Ratio Locked (`RpmPerRadS > 0`):** Gear ratio locked after `>= 10` tight-cluster IQR samples (`< 1.2%` spread); waiting for gear `N+1` ratio or top-gear WOT pull.
`N: (-)`	🔴 Red `#E74C3C`	**Unmapped / Paused:** Gear ratio not learned yet, memory reset, or game is paused / in garage menu (`IsRaceOn == 0`).

7. SIMHUB TELEMETRY PROPERTIES & ACTIONS REFERENCE (V1.0 CORE VS. V1.1 NEW)

Property `[FH6SmartShift.*]`	Ver.	Type	Description & Exact Behavior
`ShiftTriggerRpm` / `MaxUsableRpm`	v1.0	`double`	Shift Light trigger reduced by configurable driver reaction-time buffer (up to `1500 RPM` in G1–2, `600 RPM` in G3+).
`OptimalShiftRpm`	v1.0	`double`	Physical unbuffered shift target RPM (`ShiftTriggerRpm` for `(P)`, `ShiftTriggerRpm - 50` for `(L)`, `actualLimiter - 250` for `(L?)`).
`DownshiftTriggerRpm`	v1.0	`double`	Safe downshift RPM raised by braking (up to `+600 RPM`) and front dive (`+250 RPM`), capped by limiter ceiling.
`RevLimiterProximity` / `ShiftLightFlash`	v1.0	`dbl/bool`	`0.0–1.0` smooth LED progress toward buffered `ShiftTriggerRpm` / `true` when all LEDs lock at shift point.
`LiveMappingProgress` / `EngineMapStatus`	v1.0	`dbl/str`	Percentage (`0.0–100.0`) of populated active dyno bands (from `~2500–3000 RPM` to redline) / `"OK"` at `100%`, else `"LEARN"`.
`MapCoverageBar`	v1.0	`string`	20-block visual progress bar `[▓▓▓▓▓▓▓▓▒▒▒▒▒▒▒▒▒▒▒▒]` across the active powerband (refreshed every `0.5 s`).
`RpmAtMaxPower` / `RpmAtMaxTorque`	v1.0	`double`	RPM at which peak smoothed power and peak smoothed torque were recorded.
`IsDownshiftSafe` / `IsDownshiftSuggested`	v1.0	`bool`	Safe downshift flag / Downshift suggested under braking or incline load (`_climbDownshiftActive`).
`IsLearningComplete` / `IsTopGear`	v1.0	`bool`	`true` when current gear is mapped / `true` when in highest known gear (or overdrive with `dropRatio >= 0.88`).
`IsTractionGood` / `IsGripLost`	v1.0	`bool`	Dyno traction gate flag / Dashboard `SLIP` indicator (`> 1.25 G` or slip `> 0.30`).
`PowerOversteerWarning` / `OversteerSeverity`	v1.0	`bool/dbl`	Rear-axle WOT slip alert (`OVRST!`) / `0.0–1.0` rear oversteer severity scale above `15 km/h`.
`BodySlipAngle` / `AeroBottomingWarning`	v1.0	`dbl/bool`	Chassis slip angle in degrees / High-speed damper bottoming alert (`BOTTOM!` above `200 km/h`).
`Gear1Text..10Text` / `Gear1Color..10Color`	v1.0	`string`	Status labels and Hex color codes for Gears 1–10 (plus `Dbg_ChassisDive`, `Dbg_GhostLimiter`, `Dbg_SteerRaw`).
`ActiveGameMode`	v1.1	`string`	Returns `"FH6"` (Forza Horizon 6) or `"FM"` (Forza Motorsport) based on active SimHub game detection.
`CurrentCarKey` / `CurrentCarName`	v1.1	`string`	Unique vehicle/tune ID (prefixed with `FM_` in Motorsport) / User-assigned car name from `CarList.json`.
`CurrentGearColor` / `CurrentGearStatusText`	v1.1	`string`	Hex color (`#4CAF50`, `#3498DB`, `#F39C12`, `#E74C3C`) and status label of the currently engaged gear.
`LedActiveCount` / `IsShiftBeepEnabled`	v1.1	`int/bool`	`0–12` sequential LED stage count / `true` when F1 audio shift beep (`2650 Hz`) is enabled (Default: `false`).
`Dragy_Last/Best_0_100`, `100_200`, `0_200`	v1.1	`double`	Sub-frame interpolated acceleration times & Personal Bests (seconds), plus live `Dragy_Status` string.
`SlideSeverity` / `SlideTrend`	v1.1	`double`	`0.0–1.0` integrated slip level above `15 km/h` / `-1.0..+1.0` slip derivative (`IsSlideDeepening` / `Recovering`).

Property `[FH6SmartShift.*]`	Ver.	Type	Description & Exact Behavior
`SlideStateText` / `SlideBarColor`	v1.1	`string`	`"▲ DEEPENING"`, `"● HOLDING"`, `"◄ RECOVERING"`, or `"GRIP"` / Hex color (`#E74C3C`, `#FF9900`, `#2ECC71`).

⚡ SimHub Actions Reference (v1.0 Core & v1.1 New)

v1.0 `ResetPowerMemory` (resets current car) • `ResetAllCarsMemory` (full DB wipe) • `ResetLimitGears` (resets `(L)` / `(L?)` gears) • `ResetGearRatios` (resets ratios after tuning)

v1.1 `OpenDynoReport` (opens HTML Dyno, Dragy & Tuning Inspector) • `ToggleShiftBeep` (toggles F1 shift beep) • `ResetDragyRecords` (clears current car Dragy PBs).

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